

Japneet Kaur

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ACADEMIC POSITIONS	Learning Collider , Research Fellow Working with Sendhil Mullainathan, Oeindrila Dube, Ashesh Rambachan, and Peter Bergman on the Education Foundation Model Project at The Bike Shop at MIT.	July 2025 - present
EDUCATION	Massachusetts Institute of Technology, Boston Certified Course: Designing and Running Randomized Evaluations	Aug 2025 - Dec 2025
	Indira Gandhi Institute of Development Research, Mumbai Ph.D. in Economics (<i>Development Economics</i>) Thesis Title: <i>Gender Gaps in Academic Achievement - Evidence from An Emerging Economy</i> Thesis Committee: Supreet Kaur (UC Berkeley), S. Chandrasekhar & Subrata Sarkar (IGIDR).	2021 - June 2025
	Indira Gandhi Institute of Development Research, Mumbai MPhil. in Economics (<i>General Equilibrium Theory</i>) Thesis Title: <i>Competitive Equilibria and Robust Efficiency with Club Goods</i>	2019-21
	Indira Gandhi Institute of Development Research, Mumbai M.Sc in Economics	2017-19
	Panjab University, Chandigarh BA (Honors) Economics	2015-17
FIELDS	Primary: Economics of Education, Development, Labor & Behavioral Economics Secondary: General Equilibrium Theory, Game Theory, Microeconomic Theory	
JOB MARKET PAPER	<i>Lagging Boys in Academic Achievement: Is Gender Discrimination the Cause?</i> When do gender gaps in achievement emerge, how do they evolve, and what explains them? We study these questions using a unique longitudinal dataset of 117,402 students tracked annually across 117 rural schools in four North Indian states. Contrary to prevailing evidence from developing countries, we document a large and widening female advantage. Gender gaps are present as early as Grade 1 and increase steadily over time. In English, the gap increases from 0.2 standard deviations (SD) in Grade 1 to 0.7 SD by Grade 10; in mathematics, it rises from 0.1 to 0.4 SD. This pattern persists even among opposite-gender siblings and twins, suggesting that shared household and school environments cannot account for the gap. Exploiting quasi-random assignment of students to classrooms, we find limited roles for teacher gender and peer composition. Instead, non-cognitive traits—such as attentiveness and diligence—explain a substantial share of the female advantage. Tracing the origins of these behavioral differences, we uncover a novel mechanism: gender discrimination against girls induces compensatory academic effort among daughters and attenuates academic incentives for sons—a mechanism we formalize in a theoretical model.	
PAPER PUBLICATIONS	<i>Competitive Equilibria and Robust Efficiency with Club Goods</i> , with Anuj Bhowmik, Journal of Mathematical Economics 108 (2023): 102876.	
WORKING PAPERS	<i>Stable Sets in Economies with Club Goods (Under Review)</i> Economic Theory In an economy with club goods, we introduce the concept of von Neumann-Morgenstern stable sets.	

Our main result characterizes stable sets in a continuum club economy with finitely many agent types by showing they correspond exactly to the stable sets in a finite economy in which each type is represented by a single agent. The framework features a continuum of agents interacting under a decentralized, price-taking equilibrium concept. Agents trade private goods and may simultaneously hold memberships in multiple clubs, where memberships are discrete and indivisible. Unlike private goods, club goods are accessed only through joint participation: an individual either joins a viable club—one that satisfies compositional constraints—or is excluded entirely. Finally, we incorporate farsighted behavior into the analysis and show that sophisticated stable sets in utility and allocation spaces coincide.

[When Low Scores Don't Tell The Full Story-A Composite Indicator of Student Achievement](#)

This paper introduces a new composite indicator of student achievement, grounded in an axiomatic framework. Unlike conventional measures that assign equal weight to all subjects, our index applies student- and subject-specific weights, placing greater emphasis on areas where a student performs well. This allows for a more individualized assessment, recognizing strengths in non-core subjects like music, sports, or social sciences. Using test score data from 44,173 students studying in 117 private English medium schools in rural North India, we compare our indices with the traditional average score index. The results show that a substantial proportion of students initially ranked in the bottom quartile move up significantly under our metric, highlighting overlooked talent. The proposed indices CS1 and CS2 markedly increase mean scores from 0.696 (under the original index CS0) to 0.838 and nearly 1.0, respectively, while sharply reducing standard deviations from 1.88 to 0.129 and 0.017.

WORKS IN PROGRESS

Does Higher Motivation Drive Better Outcomes? Evidence from a Regression Discontinuity Design in India

In this paper, we estimate the causal impact of motivation on student learning using a Regression Discontinuity Design (RDD) centered around a salient academic threshold. Specifically, we exploit the discontinuity created by the 90-mark cutoff in Grade 10 national examinations, where students scoring 90 or above receive substantial public recognition, including the publication of their photographs in local newspapers and widespread acknowledgment from schools and the community. Students scoring just below this threshold, however, receive no such recognition. This sharp difference in public acknowledgment serves as a natural experiment to measure the effect of external motivation on subsequent academic outcomes. By tracking students' performance in the Grade 12 national examinations, we identify the causal effect of social recognition on academic achievement, isolating the role of motivation from other confounding factors.

How Effective is Online Learning? Evidence from Rural India

This paper examines the long-term impact of COVID-19-induced school closures and the shift to online instruction on student learning outcomes. Using a rich dataset of over 45,000 students from 117 rural schools in North India, we provide causal evidence on the effectiveness of online education. Although interpreting post-lockdown test scores is complicated by potential confounders, such as illness-related interruptions, technical constraints, and exam-related irregularities, we argue that the institutional and contextual setting of our study mitigates these concerns. The rural farming background of most students protected them from income or health shocks related to pandemics. Both students and teachers adapted to technological challenges over time, reducing their effect on outcomes. Examination-related biases are addressed through the use of machine-graded MCQ tests and an offline exam held post-lockdown, comparable in design and difficulty to pre-pandemic assessments. Furthermore, uniform exposure to online instruction in all schools eliminates selection bias. Leveraging

this natural experiment, we find that online education during the pandemic had a significant and measurable effect on student achievement.

AWARDS AND HONORS	Chancellor's Prize for securing the second highest GPA, M.Sc. Economics, IGIDR	2019
	Chancellor's Prize for Being the Best All-Round Student at IGIDR	2018-21
	Chancellor's Prize for securing third rank, B.A. Eco. (Hons.), Panjab University	2017
	Gold Medal in Inter College Tennis Tournament, Panjab University	2016
	Gold Medal in 100m, 200m, 4*100m relay, Panjab University Athletics Meet	2015,16
	Gold Medal in North Zone CBSE Hockey Championship	2009-13

FIELDWORK AND DATA EXPERIENCE	Learning Collider (working on Education Foundation Model at Bike Shop@MIT)	
	Designing and overseeing the implementation of a randomized controlled trial (RCT) evaluating how AI can improve student learning outcomes (ongoing).	

Indira Gandhi Institute of Development Research, Mumbai

Collected, cleaned & analyzed a large-scale longitudinal data of 116,026 students across 117 schools (2012–2024), linking test scores with parental & teacher background.

Designed & implemented detailed protocols for data cleaning, merging across waves, constructing student-level panels & standardizing variables for econometric analysis.

RESEARCH POSITIONS	Indira Gandhi Institute of Development Research, Mumbai	
	Research Assistant for Shubhro Sarkar (Microeconomic Theory)	2022-25
	Research Assistant for Subrata Sarkar (Development Economics)	2021-22
	Research Assistant for Taniya Ghosh (Macroeconomics)	2020-21

Indian Statistical Institute, Delhi

Research Intern for Bharat Ramaswami June - July 2018

J-PAL South Asia

Research Intern for Supreet Kaur and Heather Schofield
Project: Impact of computer adaptive learning on children's education July 2016

PRESENTATIONS	Econometric Society North American Summer Meeting (upcoming)	2026
	Econometric Society Africa Meeting (upcoming)	2026
	Royal Economic Society Annual Conference 2026 (upcoming)	2026
	BREW ESA Conference at Ashoka University	2024
	Winter School at Delhi School of Economics	2024

TEACHING EXPERIENCE	Indira Gandhi Institute of Development Research, Mumbai	
	Teaching Assistant for Microeconomics I (Master's level)	2022-25
	Teaching Assistant for Game Theory (Master's level)	2022-25
	Teaching Assistant for Econometrics II (Master's level)	2021-22
	Teaching Assistant for Macroeconomics II (Master's level)	2021-22
	Teaching Assistant for Special Topics in Microeconomics (Master's level)	2020-21

SOFTWARE SKILLS STATA, R, Python, MATLAB, Latex